

SEED STENCILS FOR DIFFUSION THROUGH MATERIAL EDGES TOO STEEP FOR THE GRID

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ABSTRACT. A diffusion coefficient can vary smoothly on the scale of the grid, jump across an interface, or change smoothly over a width δ smaller than the node spacing h . Through such a sub-grid edge, standard high-order stencils that sample the coefficient at the nodes are first order until the grid resolves the edge, with a knee at $h \approx \delta$, and the interface-aware stencils built for a jump, applied at the edge's centre, sit on an error floor of order δ and leave it once the grid resolves the edge. We keep the nodes and replace the polynomial basis of the stencils that see the edge by *seeds*. The seeds are the profiles of solutions polynomial in time, continued through the edge by ordinary differential equations along the normal or along the edge's own coordinates; in one dimension they are the classical formal powers of the Sturm–Liouville operator, and as $\delta \rightarrow 0$ they reduce to the interface-aware stencils of a jump. On scattered nodes the resulting RBF-FD stencils are fourth order at every δ tested, elliptic and parabolic, with no switch between regimes. We compare them with the coefficient treatments of diffusion practice. The comparison is against coefficient sampling on the same nodes and against our implementation of the cell harmonic and arithmetic means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, on one contrast and one test problem per dimension. Through an edge the grid does not resolve none of them is better than second order; the conservative scheme is exact at the one-dimensional equilibrium and, on the sinusoidal medium at $\delta = 0$, ahead of the seeds below 801 nodes, where the seeds' line is pre-asymptotic; on the scattered nodes none improves on sampling by more than a factor of 1.75 in the RMS norm, and the seeds lie 2.3 to 4.7 orders below the best of them over 1250 to 40,000 nodes.

1. INTRODUCTION

Heat conduction through a composite material is governed by

$$(1) \quad u_t = \nabla \cdot (\alpha \nabla u) \equiv Lu$$

and, at equilibrium, by $Lu = 0$, with a conductivity $\alpha > 0$ that changes sharply between materials. At the two ends of the range of media the numerical picture is settled. Where α varies smoothly on the scale of the grid, a high-order stencil samples it at the nodes and keeps its order. Where α

jumps across an interface, the temperature and the flux $\alpha \partial_n u$ are continuous while the gradient is not, a standard stencil across the interface is first order, and the remedies rebuild the stencils or the elements that the interface cuts so that they satisfy the interface conditions (Section 1.1). One of them, the interface-aware RBF-FD stencils of [Mar16, MF17b] that this paper extends, keeps fourth order on scattered nodes, elliptic and parabolic, with a few rows of nodes placed across each interface. Between the two ends lies a third case: an edge that is smooth, so that no interface condition applies, but changes over a width δ smaller than the node spacing h . Every material edge is smooth on some scale, a model that resolves an edge on a fine grid presents it as sub-grid on a coarser one, and any regularisation of a jump produces one.

A standard stencil through a sub-grid edge samples α on its two sides, sees a jump, and treats the coefficient as smooth between its nodes, which is the first-order failure of the jump case. It recovers once the grid resolves the edge. In the one-dimensional ramp problem of Section 4.2, the error of the standard fourth-order operator through a 9 : 1 edge of width $\delta = 0.01$ or 0.0025 falls by 100 to 200 times between $h = 2\delta$ and $h = \delta/2$, where a first-order line falls by four. On scattered nodes in two dimensions (Section 6.2) its elliptic error stays at 0.86 to 0.94 of its error at the jump while $h \gtrsim 6\delta$, falls to 0.081 to 0.084 of it at $h \approx \delta$ for the three widths that reach it, and converges at the fifth order of the bulk stencils below. We call the turn at $h \approx \delta$ the knee.

The interface-aware stencils do not reach this regime either. Applied with the centre of the edge as the interface and the two sides' coefficients as if δ were 0, they converge to the jump problem's solution, which is not the smooth problem's: the resistance $\int dx/\alpha$ of a tanh edge between conductivities a and b falls short of the jump's by $c\delta$, $c = (a - b) \ln(a/b)/(2ab)$, so their error sits on a floor of order δ while the grid does not resolve the edge. Once it does, the rebuilt rows keep enforcing a kink that the solution lacks, and their error grows: through a 9 : 1 edge with $\delta = 0.01$ it reaches 0.11 at 1600 nodes in the one-dimensional equilibrium of Section 4.3. On scattered nodes they beat the standard stencils only while $h \gtrsim 3\delta$, and by at most a factor of 6. Using them means knowing δ and switching them off.

This paper keeps everything about the scheme except the basis of the stencils that see the edge. In place of the monomials a stencil is made exact on *seeds*, the $t = 0$ profiles of solutions of (1) that are polynomial in time. With a constant α these profiles are the polynomials, the initial values of the heat polynomials [RW59]. In one dimension, with $L = \partial_x \alpha \partial_x$ and anchored at a stencil's evaluation point x_e , they form the chain

$$(2) \quad \phi_0 = 1, \quad \alpha \phi_1' = \alpha(x_e), \quad L\phi_k = k(k-1) \alpha(x_e) \phi_{k-2} \quad (k \geq 2),$$

with $\phi_k(x_e) = 0$ for $k \geq 1$ and $\phi_k'(x_e) = 0$ for $k \geq 2$, integrated through the edge as a first-order system of ordinary differential equations that never differentiates α (Section 3.1). The seed ϕ_1 has a constant flux, and ϕ_2 is

the profile of a solution whose rate of change u_t is constant; in general the even seeds carry $\partial_t^k u = \text{const}$ and the odd ones $\alpha \partial_x \partial_t^k u = \text{const}$, which are the interface conditions of a jump. Read across a jump, the chain is the translated polynomial basis of the interface-aware stencils (Section 3.2), so one construction covers the jump, the sub-grid edge and the resolved edge. In one dimension it is classical (Section 1.1). It is also the construction of our companion paper for the wave equation [Mar26], specialised to a single field: the two interface conditions of diffusion, on u and on its flux, are both conditions on u , so one chain serves every stencil.

In two dimensions the seeds replace the polynomial augmentation of the interface-aware RBF-FD stencils, and the Gaussian part and the saddle-point solve stay as they were. Where α varies along the normal only, as across a straight edge, seeds polynomial in the tangential coordinate turn the chain into a triangular system of ordinary differential equations in the normal coordinate, one short march per stencil (Section 5.1). Where the edge is curved or α varies along it, a profile frozen along the normal is inconsistent at $O(1)$ through an unresolved edge, and the chain is marched in the curve's own coordinates instead (Section 5.3). A resistive layer thinner than the stencil, the insulating ring of [MF17b], needs its exact width, its own series and seeds of one degree higher that carry the flux (Section 5.4). An imperfect contact costs strong-form schemes a derivative in the jump conditions, as Lombard and Piroux observed for elastic waves [LP03, LP06]; the flux seeds add the functions of one degree higher that carry the flux, and nothing else.

We measure the seeds in one and two dimensions, on elliptic and parabolic problems, against three baselines: the standard stencils with α sampled at the nodes; the interface-aware stencils of the jump applied at the edge's centre; and the coefficient treatments of diffusion practice, which change the coefficient and keep the scheme: harmonic and arithmetic means of α over a cell (over a disc on scattered nodes), the edge widened to the grid, and, in one dimension, the conservative scheme with exact face conductances. On constant pieces the one-dimensional equilibrium cannot rank these methods, since its solution has a constant flux and lies in the span of the first two seeds, on which the seeds and the exact face conductances are both exact (Section 3.5); the one-dimensional ranking is the parabolic problem's. In two dimensions the tangential variation of the solution takes it out of that span, and the equilibrium is a real test.

Contributions.

- (1) The seeds for the diffusion operator on the unchanged scattered nodes of the interface-aware RBF-FD stencils, which they reduce to as $\delta \rightarrow 0$: the one-dimensional chain (Section 3), its two-dimensional form for a straight edge in the normal coordinate, for a curved or tangentially varying edge in the curve's own coordinates, and for a resistive layer thinner than the stencil (Section 5). To our knowledge

no earlier scheme builds the basis of stencils on an unchanged scattered node set through a smooth material edge thinner than the node spacing by marching ordinary differential equations; its ingredients (the exact face conductance, the formal powers, bases from local solutions of the equation) are each classical, and in one dimension so is the construction.

- (2) In one dimension (Section 4): the knee of the standard operator, a drop of 100 to 200 times between $h = 2\delta$ and $h = \delta/2$ at a 9 : 1 edge and of 770 to 980 times at the 10 : 1 edges of the sinusoidal medium; the jump construction on its floor, the distance of order δ between the jump's solution and the smooth one, while $h \gtrsim 2\delta$ on the two-constant medium, and growing once the grid resolves the edge; and the seeds, fourth order at every δ on one line: on the ramp problem of the two-constant medium they lie within 0.1 to 0.3%, 1 to 1.5% and 3 to 5% of the jump's line at $\delta = 0.0025$, 0.01 and 0.04.
- (3) On scattered nodes through a flat band (Section 6): the seeds are fourth order at every δ tested, elliptic and parabolic, with elliptic fits of 4.23 to 4.48 over 1250 to 40,000 nodes; the elliptic errors at the five widths, the jump included, differ by factors of 1.2 to 2.4 at each node count, and at 40,000 nodes and $\delta = 0.0025$ the seeds' error is 7.99×10^{-9} against the standard operator's 8.36×10^{-5} . That is no deterioration as $\delta \rightarrow 0$ on these node sets, not a proven δ -uniform constant. The rule is to seed every row whose stencil sees an edge within 20δ , with no threshold in δ . The seeded rows' least diagonal dominance never falls below the jump-aware rows', the seeded matrices' condition estimates and GMRES counts stay bounded as $\delta \rightarrow 0$, and their spectra have no eigenvalue in the right half-plane.
- (4) Through the curved band of case 2 the tangential chain is fourth order, within 1.4 to 3.9 times of the flat seeds' RMS error from 5000 nodes at every δ (Section 6.5). On the insulating ring thinned at a fixed resistance from $s = 10^3$ to 10^{11} the seeds are exact on the matched radial profile to rounding (1.3×10^{-15} to 2.1×10^{-15}) and converge at fourth order (fits 4.43 to 4.71) at every s ; through a smooth ring whose width and edges are both below h they converge at fourth order on the nodes away from it (Section 6.6).
- (5) The comparison with the coefficient treatments (Sections 4.5 and 6.7). The comparison is against coefficient sampling on the same nodes and against our implementation of the cell harmonic and arithmetic means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, on one contrast and one test problem per dimension. Through an edge the grid does not resolve none of them is better than second order; the conservative scheme is exact at the one-dimensional equilibrium and, on the sinusoidal medium at $\delta = 0$, ahead of the seeds below

801 nodes, where the seeds’ line is pre-asymptotic; on the scattered nodes none improves on sampling by more than a factor of 1.75 in the RMS norm, and the seeds lie 2.3 to 4.7 orders below the best of them over 1250 to 40,000 nodes.

- (6) An open implementation in Python, written with the AI tools disclosed at the end of the paper, with a test for every numerical routine, in which every figure and table of this paper is drawn by a committed script from committed results files (<https://github.com/bradleypmartin/rbf-elliptic-parabolic-interfaces-2026>).

1.1. Relation to prior work. In one dimension the construction is classical in every part, and we claim nothing for it. The first seed, the constant-flux function $\alpha_e \int d\xi/\alpha$, is over one cell the exact face conductance of Tikhonov and Samarskii’s homogeneous schemes [TS62, Sam01] and, for a piecewise-constant coefficient, the harmonic-mean interface conductivity of finite-volume heat conduction [Pat80]; the conservative scheme built on it, exact at the source-free equilibrium, is second order [PXL26]. The whole chain is the representation of Sturm–Liouville solutions by recursive integrals, as power series in the spectral parameter [Kra08, KP10], taken from the anchor; for a constant coefficient its time-polynomial solutions are the heat polynomials [RW59, Wid75]. Three-point schemes of any order for $(ku)'$ with a piecewise-smooth k , whose coefficients come from Cauchy problems integrated across each cell, are the exact and truncated difference schemes [SM90, GHMK11]; three-point schemes of any order exist across a jump too [Gar89], one of them built by carrying Taylor data across the step [Suj10]. In two dimensions, local solutions of the equation that satisfy the interface conditions are the basis of the Trefftz difference schemes of FLAME, on irregular stencils too [Tsu06, Tsu10], and of the tailored finite point methods [Hua09, HZ12]; Galerkin methods build their bases from local solves [BCO94, HW97, CGH10, OZ07, MP14] and reach high order through coefficients the mesh does not resolve [Mai21, HP23], and the quasi-Trefftz spaces, one of them for the heat equation, build polynomial local solutions by Taylor expansion [IGMPS25, GPS26], which cannot reach through an edge thinner than the stencil [Mar26]. For a jump, the interface-aware RBF-FD stencils this work generalises [Mar16, MF17b] stand beside the immersed interface method and unfitted finite elements [LL94, LI06, HH02, BCH⁺15], now of fourth and sixth order [FZ20, FHM24], all of which take the jump as given. A resistive layer thinner than the grid is an imperfect interface in the limit [SP74, PP13, Has01], which Section 5.4 takes up. The standing alternative changes the coefficient rather than the stencil: harmonic or arithmetic means over a cell [Pat80, EIL01, GA92], averaging for layering finer than the grid [MDH⁺99, DDH03], or a regularised coefficient [TE03, TE06]. Those methods take the cell or the width from the grid; here the width is a property of the medium, below the node spacing, and the aim is the scheme’s full order through it. We found no prior instance of stencils

on an unchanged scattered node set whose basis is continued through a sub-grid smooth edge by ordinary differential equations, nor of such stencils reducing to a jump construction as $\delta \rightarrow 0$; the search is logged in the repository's `LITERATURE.md`. We ran the cell means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, as our implementation of each on one problem per dimension: through an edge the grid does not resolve none is better than second order, and on the scattered nodes none improves on sampling the coefficient by more than a factor of 1.75 in the root-mean-square error. We claim nothing beyond that measurement.

Outline. Section 2 fixes the problems, the discretisations and the interface-aware stencils of a jump that the seeds extend. Section 3 constructs the seeds in one dimension and proves what can be proved about them. Section 4 measures them against the standard operator, the jump construction and the coefficient treatments. Section 5 builds the seeds on scattered nodes for straight, curved and doubly sub-grid features. Section 6 measures them there, elliptic and parabolic, with the matrices they make. Section 7 records the limitations and what we would do next.

2. SETTING

Notation. The symbols the sections share, fixed here:

n, h	node count and spacing: $h = 2/(n-1)$ in one dimension and the row spacing $1/\text{round}(0.95\sqrt{n})$ in two
$\alpha; \alpha^-, \alpha^+; L$	conductivity; the pieces on the two sides of an edge; $L = \nabla \cdot (\alpha \nabla \cdot)$
$\delta; \sigma$	width of a tanh edge, $\delta = 0$ the jump; the blend weight $\sigma(z) = \frac{1}{2}(1 + \tanh z)$
s	the thinness of the ring of [MF17b], $1/s$ its width
$x_e; (x', y')$	a stencil's evaluation point (anchor); its local frame, y' normal to the edge
ϕ_k	the one-dimensional seeds (2)
$D_x, D_y; A$	first-derivative matrices; $\text{diag } \alpha$ at the nodes
naive, direct	$D_x A D_x (+D_y A D_y)$; $\alpha \nabla^2 u + \nabla \alpha \cdot \nabla u$ on one stencil
jump-aware	the interface-aware stencils of Section 2.4
the $\delta = 0$ construction	the same, applied at a smooth edge's centre
seeds	the operator whose rows that see an edge are rebuilt on seeds

2.1. The problem and its interface conditions. On a domain Ω in one or two dimensions we solve (1) with Dirichlet data on $\partial\Omega$, and its equilibrium $Lu = 0$, the elliptic problem. The conductivity $\alpha > 0$ is one smooth function on each side of every interface Γ . Across a jump the temperature and the heat flux are continuous at all times, hence so are all their time derivatives,

and $\partial_t^k u = L^k u$ turns these into conditions on spatial derivatives,

$$(3) \quad [L^k u] = 0, \quad [\alpha \partial_n L^k u] = 0 \quad \text{on } \Gamma, \quad k = 0, 1, 2, \dots,$$

where $[\cdot]$ is the difference of the two sides' limits, n the normal to Γ , and each side's L is taken with that side's α [Mar16, MF17b].

A smooth edge. With an edge width $\delta > 0$ the jump at an interface is replaced by a tanh transition between the two pieces,

$$(4) \quad \alpha = (1 - \sigma(d/\delta)) \alpha^- + \sigma(d/\delta) \alpha^+, \quad \sigma(z) = \frac{1}{2}(1 + \tanh z),$$

where d is the signed distance to the interface, $x - x_c$ in one dimension and the distance along the normal through the foot point on the curve in two, positive on the side of α^+ . The pieces keep their own variation, and $\delta = 0$ is the jump. A band between two interfaces is blended twice, from the outside piece into the inside one across the lower interface and back across the upper, so that a band wider than its edges has two independent transitions; Section 5.4 defines the smooth form of a layer thinner than its edges. Beyond 19δ from its centre an edge is its pieces to within two units in the last place.

Three regimes. For $\delta \gg h$ a stencil sees a smooth coefficient. For $\delta = 0$ it sees a jump, and (3) applies. For $0 < \delta \lesssim h$ it sees what looks to it like a jump between two of its nodes, but no interface condition applies, since α is smooth, and the smooth problem and the jump problem have different solutions. They differ at first order in δ : in one dimension the resistance $\int dx/\alpha$ of a tanh edge between constant pieces a and b falls short of the jump's by

$$(5) \quad c\delta, \quad c = \frac{(a-b)\ln(a/b)}{2ab},$$

which is 8.79δ for the 9 : 1 edge of Section 2.2 and is positive for either ordering of a and b .

Norms and orders. In one dimension an error is the relative ℓ^2 error at the nodes, $\|e\|_2/\|u\|_2$, and a rate is $\log_2$ of the ratio of consecutive errors as the node count doubles. In two dimensions an error is the root-mean-square (RMS) error over all nodes, the Dirichlet nodes included, and an order is the least-squares slope of $\log e$ against $\log h$ over the node counts named. A maximum-norm error is named as such wherever one is quoted, and every caption says which norm it shows.

2.2. One dimension. The one-dimensional problem is (1) with $L = \partial_x \alpha \partial_x$ on $[-1, 1]$ with Dirichlet ends. A grid has n equispaced nodes with both ends on nodes, so $h = 2/(n-1)$. Two media are used, each at $\delta = 0, 0.04, 0.01$ and 0.0025 :

- the *two-constant medium*, $\alpha = 1/9$ for $x < 0$ and 1 for $x > 0$, the single interface of the time-dependent test in the code behind [Mar16], which the dissertation does not report; the edge lies midway between two nodes when n is even;

- the *sinusoidal medium* of [Mar16, eq. (75)], $\alpha = 0.1 + 0.4 \sin 2\pi x$ on the layer $[0, \frac{1}{2}]$ and 1 elsewhere, whose two edges both have the contrast 10 : 1, since the sinusoid vanishes at both; both edges fall on nodes when $n - 1$ is a multiple of 4.

Each medium carries two problems: the equilibrium with $u(-1) = 1$ and $u(1) = 0$, and the *ramp problem*, which starts from $u = 0$, holds $u(1, t) = 0$, raises $u(-1, t)$ from 0 to 1 over $0 \leq t \leq 1$ by the C^∞ step $e^{-1/t}/(e^{-1/t} + e^{-1/(1-t)})$, and is read at $t = 2$. (The original code ramps with a logistic function, whose value and derivatives at $t = 0$ are small but not zero; the mismatch with the zero start at the corner would cap a fourth-order study.) Section 4.1 gives the reference solutions.

The *naive* operator is $D_x A D_x$ [Mar16, eq. (76)], with D_x the fourth-order first-derivative matrix of Fornberg’s weights [For88], one-sided at the ends, and $A = \text{diag } \alpha(x_j)$, which gives nine-point rows in the interior. It is the form in which a standard stencil sees a jump at all: a direct discretisation of $\alpha u_{xx} + \alpha' u_x$ with the coefficients sampled at the nodes is blind to the jumps, and its error on the sinusoidal medium stays at 0.22 as the grid is refined. Time stepping is the fourth-order backward differentiation formula (BD4) with $\Delta t = h$, one sparse LU factorisation per operator and the Dirichlet values imposed at every step; its three starting values come from classical Runge–Kutta sub-steps at half the stability limit.

2.3. Two dimensions. The two-dimensional problems are the three test cases of [MF17b], (1) on the unit strip $0 \leq y \leq 1$, periodic in x with period 1, with Dirichlet data on $y = 0$ and $y = 1$. The conductivity is one smooth piece in a band between two interfaces and another outside it.

- *Case 1*, a flat band $0.6 \leq y \leq 0.8$ with $\alpha = 0.2$ in it and 1 outside. With $u = \sin 2\pi x$ on $y = 1$ and $u = 0$ on $y = 0$ the equilibrium is $\sin 2\pi x v(y)$ with v piecewise exponential, and the parabolic problem is marched to $t = 0.1$ against the exact solution $e^t \sin 2\pi x w(y)$ of the same form [MF17b].
- *Case 2*, both interfaces bent to $y = 0.6 + 0.02 \sin 2\pi x$ and $y = 0.8 + 0.02 \sin 2\pi x$,¹ with $\alpha = 0.2 + 0.1 \sin 2\pi x \sin 2\pi y$ in the band and the boundary data of case 1.
- *Case 3*, a ring $0.35 - 1/s \leq r \leq 0.35$ about $(\frac{1}{2}, \frac{1}{2})$ with $\alpha = 1/(1.5s) + \sin 2\pi x \sin 2\pi y/(3s)$ in it and 1 outside, around a cooling circle $r = 0.05$ whose inside is removed, with $u = \sin 6\pi x$ on $y = 0$ and $y = 1$ and $u = 0$ on the circle. At $s = 10^3$ this is the insulating ring of [MF17b], 0.001 wide at a contrast of about 1500 : 1; the extremizing sweep of that paper runs s up to 10^{11} , thinning the ring at a fixed resistance, its width over its conductivity, of $1.5/(1 + \frac{1}{2} \sin 2\pi x \sin 2\pi y)$.

¹The published texts do not state the curves; these are the ones of the original code.

With $\delta > 0$ each interface of cases 1 and 2 becomes an edge (4) in the signed normal distance to its curve; the ring's smooth form is Section 5.4's. On case 1 the medium is then the one-dimensional one in y , and the reference solutions stay separable (Section 6.1).

Node sets. A set of n nodes is built as in [MF17b]. Rows hold $m = \text{round}(0.95\sqrt{n})$ nodes per unit length, so their spacing is $h = 1/m$. Six fixed rows straddle each interface, three on each side at normal distances $(\frac{1}{2} + j\frac{\sqrt{3}}{2})h$, $j = 0, 1, 2$, the middle row on each side staggered by half a spacing along the curve, so that the rows cross the interface in a hexagonal pattern; a fixed row lies on each Dirichlet curve; and the remaining nodes are placed by 100 steps of a simulated repulsion from a random start, kept clear of the straddling rows. The ring is thinner than h at every node count used, so one set of six rows straddles its midline, and both of its interfaces pass between the innermost pair. Unless a spread over node sets is quoted, each node count has one node set, from a fixed random seed.

RBF-FD. RBF-FD [FF15a, FF15b] builds each row of a differentiation matrix from a stencil. For a stencil of nodes $x_1, \dots, x_{N_s}$ about an evaluation node x_c , the weights w of a linear operator $\mathcal{L}$ solve

$$(6) \quad \begin{bmatrix} \Phi & P \\ P^\top & 0 \end{bmatrix} \begin{bmatrix} w \\ w_* \end{bmatrix} = \begin{bmatrix} \mathcal{L} \varphi(\|x - x_i\|)|_{x_c} \\ \mathcal{L} p_j(x)|_{x_c} \end{bmatrix},$$

with $\Phi_{il} = \varphi(\|x_i - x_l\|)$ for the Gaussian $\varphi(r) = e^{-(\varepsilon r)^2}$, $P_{ij} = p_j(x_i)$ over the bivariate monomials p_j up to a degree p in coordinates scaled by the stencil's radius, and w_* discarded. The constraint rows make the weights exact on the polynomials, which take over from the Gaussians under refinement [FFBB16, BFFB17]; the shape parameter is $\varepsilon = 0.4/d$, with d the distance from x_c to its nearest neighbour. Stencils have 42 nodes and degree 5 in the bulk, and 30 nodes and degree 4 within $3/\sqrt{n}$ of a Dirichlet curve and wherever a 42-node stencil would reach across an interface. Of the latter, the interface-aware construction rebuilds the rows whose 30 nodes cross an interface (Section 2.4); the seeds rebuild the rows whose stencils see an edge (Section 5.2).

Operators and solvers. The naive operator is $D_x A D_x + D_y A D_y$, the two-dimensional form of [Mar16, eq. (76)], with every node's rows of D_x and D_y built on its own stencil, so that it reaches the neighbours of the neighbours. The direct operator applies $\alpha \nabla^2 u + \nabla \alpha \cdot \nabla u$ at each node through one stencil; it is blind to a jump and serves as the control where the grid resolves the edge. Dirichlet rows are the identity. The elliptic systems are solved directly with SuperLU; the iterative study applies GMRES and BiCGSTAB to the interior system, with and without an incomplete LU factorisation and the diagonal-dominance preconditioner of [Mar16, App. B]. The parabolic problem of case 1 is marched by BD4 with $\Delta t = h$ from the exact solution's history, 3 to 19 steps to $t = 0.1$.

2.4. Interface-aware stencils for a jump. The seeds extend the interface-aware stencils of [Mar16, MF17b], whose wave-equation form is in [MFSC15, MF17a]. We summarise them in the form the later sections use; the derivations are in those references.

One dimension. Expand u on each side of an interface x_c in a polynomial of degree four about x_c , with the Taylor coefficients of that side's α entering through multiplication matrices on the coefficient vectors, and impose (3) on $L^k u$ for $k = 0, 1, 2$ and on the flux for $k = 0, 1$. These five conditions give a square *continuity matrix* per side, $C^- a^- = C^+ a^+$ on the coefficient vectors $a^\pm$. Keeping the monomials on the evaluation node's side and carrying them across with $(C^+)^{-1} C^-$ gives the *translated basis*: piecewise polynomials, one per monomial, that satisfy the five conditions. The weights of a five-node stencil that straddles x_c reproduce L at the evaluation node on this basis, a 5×5 solve per stencil, and a stencil that straddles both edges of a thin layer translates twice.

Two dimensions. A stencil whose 30 nodes lie in more than one region is given a local frame at the point of each interface nearest its centre, x' along the tangent and y' along the normal, in units of the stencil's radius, and the interface is expanded there as $y' = f(x')$. The continuity conditions equate, along the interface, the first $5 - 2k$ coefficients in x' of $L^k u$ and the first $4 - 2k$ of the normal flux $\alpha(\partial_{y'} - f' \partial_{x'}) L^k u$, fifteen conditions for degree four, one per bivariate monomial; the curvature enters through f , in the curve along which the coefficients are read and in the normal derivative. The translated basis replaces the monomials p_j in (6). The Gaussians are *warped*: in the frame of the evaluation node's side, the normal coordinate of every node is replaced by a piecewise-linear one whose slope changes by the factor α^-/α^+ across each interface, so that a Gaussian's value and its $\alpha \partial_n$ are both continuous there and the RBF part of the stencil satisfies the flux condition to first order.

This construction, with warped Gaussians, is what we call *jump-aware*. Applied with the centre of a smooth edge as the interface and the pieces' Taylor data as if δ were 0, it is the $\delta = 0$ *construction* of Sections 4.3 and 6.2. It is also the $\delta \rightarrow 0$ limit of the seeds, which reproduce it at $\delta = 0$ (Sections 3.2 and 5.2).

2.5. Scope of the implementation. The experiments run on a reimplementation of the 2016–2017 methods in Python (NumPy and SciPy, with a test for every numerical routine), written for this study. Every published figure of the one- and two-dimensional heat problems of [Mar16, MF17b] but the cornered interface's has a counterpart in it, and every published convergence line is reproduced at its published rate or better. Where the numbers part company the repository's port notes say so. Two differences bear on this paper: the iterative solvers do not break down on the ring here (BiCGSTAB converges at every node count), and the extremizing sweep does not lose convergence at $s = 10^{11}$ (every s is fourth order down to its

reference's own error). The 2016 two-dimensional code is not preserved, so neither difference could be traced to it.

The implementation leaves out the cornered interface of [Mar16, Sec. 5.4.5], which is approximated there by circular segments at second order. Its two-dimensional media are a band between two curves, periodic in x , so no edge meets the domain's boundary. The bulk stencils are degree five and the stencils that see an edge degree four, seeded or interface-aware. Each case has one contrast: 9 : 1 and 10 : 1 in one dimension, 5 : 1 on cases 1 and 2, and on the ring about 1500 : 1 at $s = 10^3$, growing with s at a fixed resistance. Every smooth edge is a tanh. Section 7 returns to these restrictions.

3. SEEDS FOR THE DIFFUSION OPERATOR IN ONE DIMENSION

Fix a stencil of the grid of Section 2.2: five nodes $x_1 < \dots < x_5$ and an evaluation point x_e , one of them. The standard weights are exact on the monomials $(x - x_e)^k$, $k \leq 4$, and the interface-aware weights of Section 2.4 on the translated basis. This section builds the basis for a smooth edge, shows that the translated basis is its limit as $\delta \rightarrow 0$, proves that every stencil solve is nonsingular, and says what the truncation error is. Each ingredient is classical in one dimension (Section 1.1), and the construction is that of our companion paper for the wave equation [Mar26], specialised to the diffusion operator.

3.1. Seeds as the profiles of solutions polynomial in time. Look for solutions of $u_t = Lu$, $L = \partial_x \alpha \partial_x$, that are polynomial in time, $u = \sum_{j=0}^J t^j g_j(x)$. Matching powers of t gives $(j+1)g_{j+1} = Lg_j$ and $Lg_J = 0$, so the profile at $t = 0$ satisfies $L^{J+1}g_0 = 0$. The heat equation spends one power of L on each power of t where the wave equation spends two, but the profiles fill the same nested kernels of L . With a constant α they are the polynomials: the time-polynomial solutions of $u_t = \alpha u_{xx}$ are the heat polynomials

$$v_k(x, t) = \sum_{0 \leq j \leq k/2} \frac{k!}{j!(k-2j)!} x^{k-2j} (\alpha t)^j$$

[RW59, Wid75], whose profiles at $t = 0$ are the monomials. The seeds are the same profiles for a variable α , anchored at the evaluation point.

Definition 3.1 (Seeds). Let $\alpha > 0$ and $\alpha_e = \alpha(x_e)$. The *seeds* at x_e are the functions $\phi_0, \phi_1, \dots$ of the chain (2): $\phi_0 = 1$, $\alpha \phi'_1 = \alpha_e$ and $L\phi_k = k(k-1)\alpha_e \phi_{k-2}$ for $k \geq 2$, with $\phi_k(x_e) = 0$ for $k \geq 1$ and $\phi'_k(x_e) = 0$ for $k \geq 2$.

Lemma 3.2. *For every k the function*

$$(7) \quad u_k(x, t) = \sum_{0 \leq j \leq k/2} \frac{k!}{j!(k-2j)!} (\alpha_e t)^j \phi_{k-2j}(x)$$

solves $u_t = Lu$, and $u_k(\cdot, 0) = \phi_k$. With a constant α , $\phi_k = (x - x_e)^k$ and $u_k(x, t) = v_k(x - x_e, t)$.

Proof. Write $c_j = k!/(j!(k-2j)!)$. By the chain, $Lu_k = \sum_j c_j (k-2j)(k-2j-1) \alpha_e^{j+1} t^j \phi_{k-2j-2}$ and $\partial_t u_k = \sum_j (j+1) c_{j+1} \alpha_e^{j+1} t^j \phi_{k-2j-2}$, and the two agree because $(j+1) c_{j+1} = c_j (k-2j)(k-2j-1)$. With α constant, $L(x - x_e)^k = \alpha k(k-1)(x - x_e)^{k-2}$ and the anchor conditions fix the two constants of integration, so $\phi_k = (x - x_e)^k$ by induction; with $\alpha_e = \alpha$, (7) is then $v_k(x - x_e, t)$. $\square$

Read one at a time, the seeds say what each monomial becomes through the edge. The seed ϕ_1 solves $L\phi_1 = 0$: it is the equilibrium of constant flux, $\phi_1(x) = \alpha_e \int_{x_e}^x d\xi/\alpha(\xi)$, and what stays smooth through the edge is its flux $\alpha\phi'_1$, not its slope. The seed ϕ_2 is the profile of $u_2 = \phi_2 + 2\alpha_e t$, whose rate of change is constant, $u_t = 2\alpha_e$; the constant matters, since $u_t = 0$ would only return $\ker L$. In general, by (7), $\partial_t^m u_{2m} = (2m)! \alpha_e^m$ is constant and $\partial_t^m u_{2m+1} = (2m+1)! \alpha_e^m \phi_1$ has a constant flux: the even seeds carry $\partial_t^m u = \text{const}$ and the odd ones $\alpha \partial_x \partial_t^m u = \text{const}$. Taken in the order $\phi_0, \dots, \phi_4$ these are the five conditions of (3) from which the translated basis of a jump is built, on u , its flux, $u_t = Lu$, the flux of u_t , and $u_{tt} = L^2 u$. In the wave equation the same alternation passes between two fields; in diffusion both kinds of condition bear on u , so one chain serves every stencil.

In one dimension the seeds are a classical object. Integrating the chain twice from the anchor gives $\phi_k(x) = k(k-1) \alpha_e \int_{x_e}^x \alpha(s)^{-1} \int_{x_e}^s \phi_{k-2}(r) dr ds$, and unrolling the recursion,

$$(8) \quad \phi_k = k! \alpha_e^{\lceil k/2 \rceil} I_k, \quad I_k(x) = \int_{x_e}^x \frac{ds_1}{\alpha(s_1)} \int_{x_e}^{s_1} ds_2 \int_{x_e}^{s_2} \frac{ds_3}{\alpha(s_3)} \cdots,$$

where $I_0 = 1$ and I_k has k nested integrals whose weights alternate between $1/\alpha$ and 1, starting from $1/\alpha$ outermost. These are the recursive integrals, $X^{(k)}$ for odd k and $\tilde{X}^{(k)}$ for even k , by which Kravchenko and Porter write the solutions of $(pu')' + qu = \lambda ru$ as power series in the spectral parameter [Kra08, KP10], here with $q = 0$, $r = 1$, $p = \alpha$, the particular solution $u_0 = 1$ and the integrals taken from the anchor. The seeds are thus the formal powers of the Sturm–Liouville operator L anchored at the evaluation point, and the coefficients of those series are the time-polynomial profiles, with λ in the role of ∂_t . The repository’s tests check the identity against the ODE march below to a relative 10^{-9} through a tanh edge.

The weights w_i of a stencil approximate Lu at x_e from the nodal values by reproducing L on the first five seeds,

$$(9) \quad \sum_{i=1}^5 w_i \phi_k(x_i) = (L\phi_k)(x_e) = \begin{cases} 2\alpha_e, & k = 2, \\ 0, & k = 0, 1, 3, 4, \end{cases}$$

since $L\phi_k = k(k-1) \alpha_e \phi_{k-2}$ vanishes at x_e for every k but 2. These are the conditions the interface-aware stencils impose on the translated basis, and

with a constant α they return α times the standard second-derivative weights [For88], which the implementation reproduces to rounding. Stencils exact on a local space other than the polynomials go back to fitted operators and to HODIE [AS55, Il'69, LR80, RST08]. Three-point schemes whose coefficients come from Cauchy problems integrated across a piecewise-smooth coefficient are the exact and truncated schemes [SM90, GHMK11], and the seed stencils are a five-point variant of them.

First-order form. The chain is integrated as a first-order system in each seed and its flux $\psi_k = \alpha \phi'_k$,

$$(10) \quad \phi'_k = \psi_k / \alpha, \quad \psi'_k = k(k-1) \alpha_e \phi_{k-2},$$

with $\phi_0(x_e) = 1$, $\psi_1(x_e) = \alpha_e$ and every other initial value zero, so that α is never differentiated. The ten states march together from x_e outward to each side of the stencil. Across a jump ϕ and ψ are continuous, which is L read in the weak sense: the march stops at the jump and reads α on each side from the piece that owns it, so a jump is marched like any other width, and a stencil that straddles both edges of a thin layer marches through both.

Stencil coordinate. In the coordinate $\xi = (x - x_e)/h_s$, with $h_s = \max_i |x_i - x_e|$, the functions $\phi_k(x)/h_s^k$ satisfy the same chain with $\alpha(x_e + h_s \xi)$ in place of α . They are close to ξ^k , and the matrix $A_{ki} = \phi_k(x_i)/h_s^k$ of (9) is a perturbed Vandermonde matrix; the weights are $w = \tilde{w}/h_s^2$, with $A\tilde{w}$ equal to the right-hand side of (9). For the centred stencil, $\xi_i = 0, \pm\frac{1}{2}, \pm 1$, the condition number of A is 23.5 with a constant α . On the stencil of Figure 1, half a cell from a 9 : 1 edge, it is 90 at $\delta = 0$ and 138 at $\delta = h/2$, and 24 again once $\delta \geq 10h$, where the seeds are close to the monomials.

The march and the seeded rows. The march is an explicit adaptive Runge–Kutta method of order eight (SciPy’s DOP853) at relative tolerance 10^{-13} and absolute tolerance 10^{-15} , with segment boundaries at each edge centre and at 10δ on either side of it, so that the step never has to find the edge inside a long step over flat material. A row is seeded when the span of its stencil meets $(x_c - 20\delta, x_c + 20\delta)$ for an edge centre x_c , beyond which the edge is its pieces to the last bit; every other row is the standard one. At $\delta = 0$ the seeded rows are exactly those that straddle the jump; at $\delta = 0.0025$ and 0.04 on 100 nodes they number 8 and 84; once 20δ covers the domain, every row is seeded. No threshold in δ switches the seeds on or off: a resolved edge costs them nothing (Section 4.4), and at $\delta = 0$ they are the jump’s weights (Section 3.2). Rows whose nodes sit at the same offsets from x_e march as one system, so an operator costs some sixteen calls of the integrator whatever the number of seeded rows: 0.11 s for the 1284 seeded rows of $\delta = 0.04$ on 1600 nodes. The march has one floor of its own. With L_h the discrete operator, the seeded rows’ residual on the exact equilibrium, $h^2 \max |L_h u|$, is at most 2.5×10^{-16} at 200 nodes while the march crosses the edge in a few steps ($\delta \geq h/10$), and 2.4×10^{-13} and 5.8×10^{-13} at $\delta = h/100$ and $h/1000$, where it takes many. The equilibrium solution shows it once

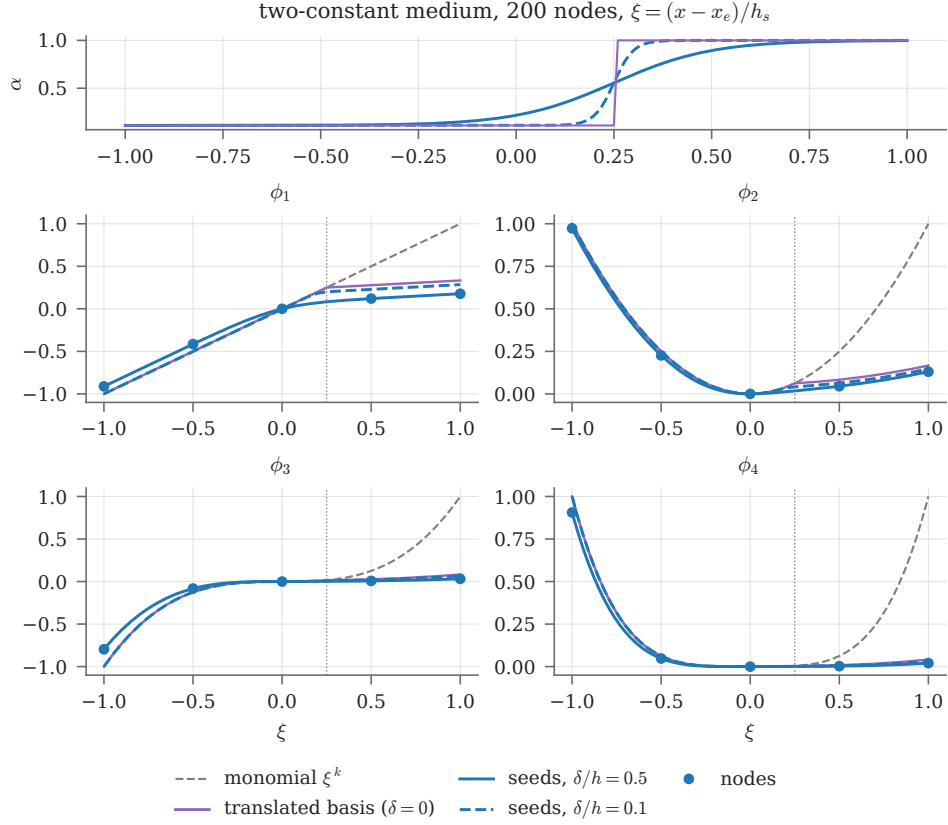


FIGURE 1. The seeds $\phi_1, \dots, \phi_4$ of one five-point stencil of the two-constant medium at 200 nodes, whose evaluation point x_e lies half a cell to the left of the 9 : 1 edge (dotted), in the stencil coordinate $\xi = (x - x_e)/h_s$ with $h_s = 2h$; α across the stencil on top. Grey, dashed: the monomials ξ^k . Purple: the translated basis of the jump ($\delta = 0$). Blue: the seeds at $\delta = h/2$ (solid, with their values at the nodes) and $\delta = h/10$ (dashed). Each seed is multiplied by $(\alpha_0/\alpha_e)^{\lceil k/2 \rceil}$, with α_0 the jump's value at x_e and α_e the edge's, so that the lines compare across δ ; the factor is the normalisation of (2), which the weights never see. On the node's side of the jump its seeds are the monomials, and across it the slope of ϕ_1 falls ninefold; the seeds of a smooth edge bend through it instead of kinking, and approach the translated basis at first order in δ/h .

$\delta \lesssim h/40$, at 10^{-11} to 4×10^{-11} . A collocation integration of the same chain would remove the floor; we have not built one.

3.2. Nested kernels and the jump as the limit. Since L sends each seed two steps down the chain, $\text{span}\{\phi_0, \dots, \phi_k\}$ is the $(k+1)$ -th space of

$$(11) \quad \{1\} \subset \ker L \subset \ker \partial_x L \subset \ker L^2 \subset \ker \partial_x L^2 \subset \dots,$$

where $\ker \partial_x L^m$ holds the g with $L^m g$ constant. The spaces depend on neither the anchor nor the normalisation, so the weights do not either; only the basis does.

For a jump, read L in the weak sense, ϕ and $\alpha\phi'$ continuous across it. Then $\ker L$ is spanned by 1 and the piecewise-linear function whose slope changes by the factor α^-/α^+ across the jump, the translated x of Section 2.4. On constant pieces $L\phi_2 = 2\alpha_e$ makes $\phi_2'' = 2\alpha_e/\alpha^\pm$ on the two sides, with ϕ_2 and its flux continuous, which is the continuity of u_t and of its flux; and so on up the chain, each condition $\partial_t^m u = \text{const}$ or $\alpha \partial_x \partial_t^m u = \text{const}$ being one of (3). For constant pieces the chain across a jump is therefore the translated basis, the anchor conditions picking one basis of its span, and the interface-aware stencils are the seed stencils at $\delta = 0$: one construction with two implementations, algebra for a jump and an ODE march for a smooth edge. On the stencil of Figure 1 the march reproduces the translated basis to 3×10^{-15} function by function, and its weights to 7×10^{-16} . The same device, Taylor data carried across a step to make the weights, appears in photonics [Suj10], and three-point schemes of any prescribed order across a jump in the coefficient are Gartland's [Gar89].

As $\delta \rightarrow 0$ the seed weights approach the jump's at first order in δ/h . On the stencil of Figure 1, $\max_i |w_i - w_i^{\text{jump}}| / \max_i |w_i^{\text{jump}}|$ is 84, 46, 11, 1.1 and 0.11 percent at $\delta/h = 1, \frac{1}{2}, 0.1, 0.01$ and 0.001, and the seeds approach the translated basis at the same rate. A stencil that straddles both edges of a layer two cells thick, $\alpha = 1 | 0.1 | 1$, marches through both: at $\delta = 0$ its weights are those of the jump construction translated twice, to 7.6×10^{-15} , and at $\delta/h = 0.1, 0.01$ and 0.001 they differ from them by 22, 1.9 and 0.18 percent.

When the pieces vary the limit is not exact. The interface-aware construction expands each side's α to degree four about the interface and truncates the operator there (Section 2.4), while the seed ϕ_2 solves $(\alpha\phi_2')' = 2\alpha_e$ exactly, $\phi_2 = (x - x_e)^2 - \frac{2}{3} (\alpha'(x_e)/\alpha_e) (x - x_e)^3 + \dots$, so that within a stencil the two five-dimensional spaces differ at relative order $h\alpha'/\alpha$. On the sinusoidal medium, whose layer has $\alpha'/\alpha = 25$ at its edge, the seed and jump weights of the row whose evaluation point is that edge differ at $h = 0.01$ by 1.25 times the largest weight at $\delta = 0$ and by 1.26 to 2.24 times for δ/h from 0.001 to 1, and the difference falls with h . Neither space is wrong, each being exact on its own five functions; on this medium the seed operator's error is the smaller, by 6 to 44 times at equilibrium and 6 to 38 times on the ramp problem at $\delta = 0$ over 101 to 1601 nodes. There the seeds are measured against the reference solution, never against the jump construction.

3.3. Well-posedness. The stencil solve is nonsingular for every edge width, the jump included, and for every set of distinct nodes, not only the ones we used.

Proposition 3.3. *Let α be positive and bounded away from zero on an interval I containing x_e , and continuous on I except at finitely many jumps, across which L is read in the weak sense. Then for every k the seeds $\phi_0, \dots, \phi_k$ form a complete Chebyshev system on I : interpolation in their span at any $k + 1$ distinct points of I is uniquely solvable. Where α is smooth, as through a tanh edge between smooth pieces, they form an extended complete Chebyshev system. In particular (9) has a unique solution for every $\delta \geq 0$ and every five distinct nodes in I .*

Proof. By (8) the seeds are nonzero multiples of $I_0, \dots, I_k$, nested integrals whose weights $v_1 = 1/\alpha$, $v_2 = 1$, $v_3 = 1/\alpha, \dots$ (outermost first) are positive. Such a family is the canonical basis of the kernel of the operator in Pólya form $\partial_x \frac{1}{v_k} \partial_x \dots \frac{1}{v_2} \partial_x \frac{1}{v_1} \partial_x$, which is $\partial_x L^m$ for $k = 2m$ and L^{m+1} for $k = 2m + 1$, and whose kernel is the $(k + 1)$ -th space of (11). Suppose $g = \sum_{j \leq k} c_j I_j$ vanishes at $k + 1$ distinct points of I . Then $g' = v_1 g_1$, where $g_1 = \sum_{1 \leq j \leq k} c_j I'_{j-1}$ is a combination of the k nested integrals $I'_0, \dots, I'_{k-1}$ with the weights $v_2, v_3, \dots$, and g_1 is continuous. Between two consecutive zeros $a < b$ of g , $\int_a^b v_1 g_1 = 0$ with $v_1 > 0$, so g_1 vanishes in (a, b) : it has k distinct zeros. By induction on k , $g_1 \equiv 0$, so $c_1 = \dots = c_k = 0$, and then $g = c_0$ vanishes, so $c_0 = 0$. Only the positivity and integrability of the weights entered, so the argument holds across finitely many jumps. Where α is smooth the operator is disconjugate on I and the family is an extended complete Chebyshev system [P622, KS66, Cop71]. $\square$

When x_e lies on a jump, as it does for the rows whose evaluation node falls on an edge of the sinusoidal medium, α_e is the value of the piece that owns x_e ; ϕ_1 then has slope 1 on that side and α_e/α on the other, and nothing else changes, since α_e enters the seeds only as a factor. The argument is the one the companion paper makes for the two fields of the wave equation [Mar26], shortened by (8).

3.4. Truncation error. Interpolation in an extended complete Chebyshev system has a Newton form whose remainder carries the operator that annihilates the space where the polynomial remainder carries the $(k + 1)$ -th derivative [Mü73]. For the five seeds that operator is $M = \partial_x L^2$, and the local truncation error of a seed stencil on a smooth function g is

$$(12) \quad \tau(g) = \sum_{i=1}^5 w_i g(x_i) - (Lg)(x_e) = O(h^3 M g),$$

with a constant that the remainder expresses through $1/\alpha$ inside the stencil. That is third order, not fourth: a second derivative on five points is generically third order, and the centred standard stencil gains its fourth

only because its h^3 term vanishes by symmetry, which an edge breaks. The interface-aware rows of a jump show the same. On the function of degree five in their translated basis, which lies outside the space they reproduce, the h^3 term is 0.32, 0.084 and 172 (in the stencil coordinate) for a 9 : 1 jump half a cell and one and a half cells from the evaluation node and at the node itself, and zero to rounding for equal pieces.

The global order is still four. A local error of order h^3 on a fixed number of rows costs one order less in the solution than the same error on every row, since the inverse of the discrete operator has entries of order h , so the solution error is $O(h^4)$. This is what the interface-aware stencils reach at a jump (rates 3.85 to 4.19, elliptic and parabolic, in our port of [Mar16]), and it is the phenomenon Beale and Layton prove for interface corrections, where an $O(h)$ truncation error at the interface leaves the solution second order [BL06]. The bound (12) has two readings, and Section 4 is built on them.

Equilibrium. If $Lu = 0$ then $Mu = 0$, and more than that: u has a constant flux, so on every seeded stencil it lies in $\text{span}\{\phi_0, \phi_1\}$, and (9) reproduces Lu there exactly, at any δ and any h (Section 3.5).

Time dependence. If $u_t = Lu$ then $L^2u = u_{tt}$ and $Mu = \partial_x u_{tt}$, the x -derivative of another solution of the heat equation. Since u_{tt} has a continuous flux, $\partial_x u_{tt}$ is bounded independently of δ and changes across the edge by the ratio of the pieces, whatever the width. The standard stencil's error is driven by $u^{(5)}$ instead, which grows like δ^{-4} inside the edge, and the rows of $D_x A D_x$ that see an unresolved edge miss a fixed fraction of the jump in slope, an error of order $1/h$ in Lu . The bound does not make the constant in (12) independent of δ/h , since the remainder involves $1/\alpha$ across the stencil. Measured on a separable solution $e^{ct}v(x)$ of the two-constant medium from 100 to 400 nodes, the seeded rows' residual is third order (rates 3.06 to 3.78), its constants at $\delta = 0.01$ and 0.0025 within a factor of two of each other, and the standard rows' residual is fourth order (3.94 to 4.01).

Diffusion smooths, so for $t > 0$ every $\partial_t^k u$ is itself a solution and the second reading holds for the whole chain. And since $\partial_x(\alpha u_x) = u_t$, the flux changes across an edge of width δ only by $\int u_t dx = O(\delta)$: in diffusion a sub-grid edge is close to a jump at all times. That is why the $\delta = 0$ construction is a serious baseline here (Section 4.3), and why the seeds' margin over it is set by δ .

3.5. The elliptic problem in one dimension. The one-dimensional equilibrium has a constant flux $\alpha u' = B$, so $u = u(-1) + B \int_{-1}^x d\xi/\alpha$, and two constructions are exact on it at every h and every δ when the pieces are constant.

The first is the seed operator. Its seeded rows are exact by the first reading of Section 3.4, and its standard rows see a linear function on a constant piece. At $\delta = 0$ it is the interface-aware operator of the jump.

The second is the conservative three-point scheme with the face conductances

$$(13) \quad a_{i+1/2} = h / \int_{x_i}^{x_{i+1}} \frac{d\xi}{\alpha(\xi)},$$

whose discrete flux $a_{i+1/2}(u_{i+1} - u_i)/h$ equals B at the exact nodal values whatever α is, so that the exact solution satisfies every discrete equation. It is the homogeneous scheme of Tikhonov and Samarskii [TS62, Sam01], whose face coefficient is the conductance of the constant-flux seed over one cell. For a piecewise-constant α it is the harmonic-mean interface conductivity of finite-volume heat conduction, which Patankar derives from the same steady one-dimensional flux [Pat80, §4.2-3] and which is the exact interblock transmissivity of groundwater models for steady one-dimensional flow through blocks of uniform transmissivity [GA92]. Pan, Xu and Li call the scheme the harmonic average method and prove it second order [PXL26]. The exactness belongs to the conservative form: the same harmonic mean sampled at the nodes and used as A in $D_x A D_x$, the like-for-like changed medium of Section 4.5, leaves a residual above 0.5 on the exact equilibrium of the 9 : 1 jump at 100 nodes, whether the mean is taken over one cell or two.

So on constant pieces the one-dimensional equilibrium cannot rank the seeds against the conservative scheme. It is the sanity check and the limit case, and the one-dimensional ranking is the parabolic problem's, where the conservative scheme is second order and the seeds are fourth (Section 4.5). Both exactness results are classical, and so is the conservative scheme's second order; we observe them and claim none of them. Where the pieces vary, as in the sinusoidal medium, the equilibrium does rank the two, for a reason that has nothing to do with the edge: the conservative scheme stays exact, while the seed operator's standard rows inside the layer keep the error of a smooth coefficient, and it is exact only where the seeded rows cover the layer (Section 4.4).

4. ONE-DIMENSIONAL RESULTS

This section measures the naive operator, the $\delta = 0$ construction and the seeds, and then the coefficient treatments, on the two media of Section 2.2 at $\delta = 0, 0.04, 0.01$ and 0.0025 . Node counts double from 50 to 1600 on the two-constant medium and from 101 to 1601 on the sinusoidal one, whose edges then fall on nodes; $h = \delta$ falls at $n = 51, 201$ and 801 for the three widths. Errors are relative ℓ^2 errors at the nodes and rates are per doubling of n (Section 2.1). The ramp problem is marched by BD4 with $\Delta t = h$, so its errors are those of the fully discrete scheme; Section 4.4 separates the time integrator's share.

4.1. Media and references. With $\delta > 0$ each medium blends its pieces across each edge by (4), the two edges of the sinusoidal medium in turn, so

that the pieces' weights remain a partition of unity; at $\delta = 0$ the blend is the jump to the last bit. Both references cut $[-1, 1]$ into elements on which α is smooth up to their ends: the pieces for a jump, and for an edge also the points $x_c \pm m\delta$, $m = 1, 3, 9, 27$, so that outside the core $|x - x_c| \leq \delta$ each element spans at most a factor of three in distance from the centre and is resolved by a polynomial of modest degree.

The equilibrium reference is $u = u(-1) + B \int_{-1}^x d\xi/\alpha$, with B fixed by $u(1)$ and the integral taken by Gauss quadrature on the elements. Through a tanh edge between constants a and b the integrand has the antiderivative

$$\delta \left[\frac{z}{a} + \frac{a-b}{2ab} \ln(a + b e^{2z}) \right], \quad z = \frac{x - x_c}{\delta},$$

which the quadrature reproduces to 10^{-14} ; subtracting the jump's z/a and z/b on the two sides leaves the deficit (5) as $z \rightarrow \pm\infty$. The parabolic reference is Chebyshev collocation on the same elements, 20 points on each and elements wider than 0.1 split, with u and αu_x matched at every element end, integrated to $t = 2$ by SciPy's implicit Radau method at relative and absolute tolerances 10^{-9} and 10^{-11} ; the absolute tolerance sits above the round-off of the collocation system on elements of width of order δ , which a tighter tolerance would make the step control chase. Against a second resolution, 24 points and elements to 0.05, it agrees to 5×10^{-12} on the two-constant medium and 6×10^{-11} on the sinusoidal one, at worst over the four widths. That is round-off, not truncation, and the finest errors of the seeds sit on it (Section 4.4).

Against each $\delta > 0$ line the figures also draw the *floor* $\|u_0 - u_\delta\|_2 / \|u_\delta\|_2$ at the nodes, the distance between the jump problem's solution u_0 and the smooth one u_δ , from the two references. It is what a method that converges to the jump's solution reaches (Section 4.3).

4.2. The knee. Figure 2 gives the three operators at every width, and Table 1 the naive operator's and the construction's errors on the ramp problem. At $\delta = 0$ the naive operator is first order, with rates of 0.98 to 1.02 on the two-constant medium and 1.00 to 1.01 on the sinusoidal one from 201 nodes. While $h \geq 4\delta$ its lines at $\delta > 0$ stay with the jump's, at 0.43 to 1.05 times its error and at rates of 0.9 to 1.6 that depend on where the nodes happen to sample the edge: its rows that see the edge treat it as a jump between two of their nodes, the first-order failure of Section 3.4.

The knee is at $h \approx \delta$. Between $h = 2\delta$ and $h = \delta/2$ the ramp problem's error falls by 117 and 199 times on the two-constant medium at $\delta = 0.01$ and 0.0025, and by 771 and 977 times on the sinusoidal medium, where a first-order line falls by four. On the two-constant medium, whose edge lies midway between two nodes, most of the fall comes in the one doubling from $h = \delta$ to $\delta/2$, at rates of 5.9 to 6.8; on the sinusoidal medium, whose edges lie on nodes, it comes a doubling earlier, at 5.2 to 5.4. From $h = \delta/2$ on, the nodes resolve the tanh and the lines are fourth order, at rates of 3.95

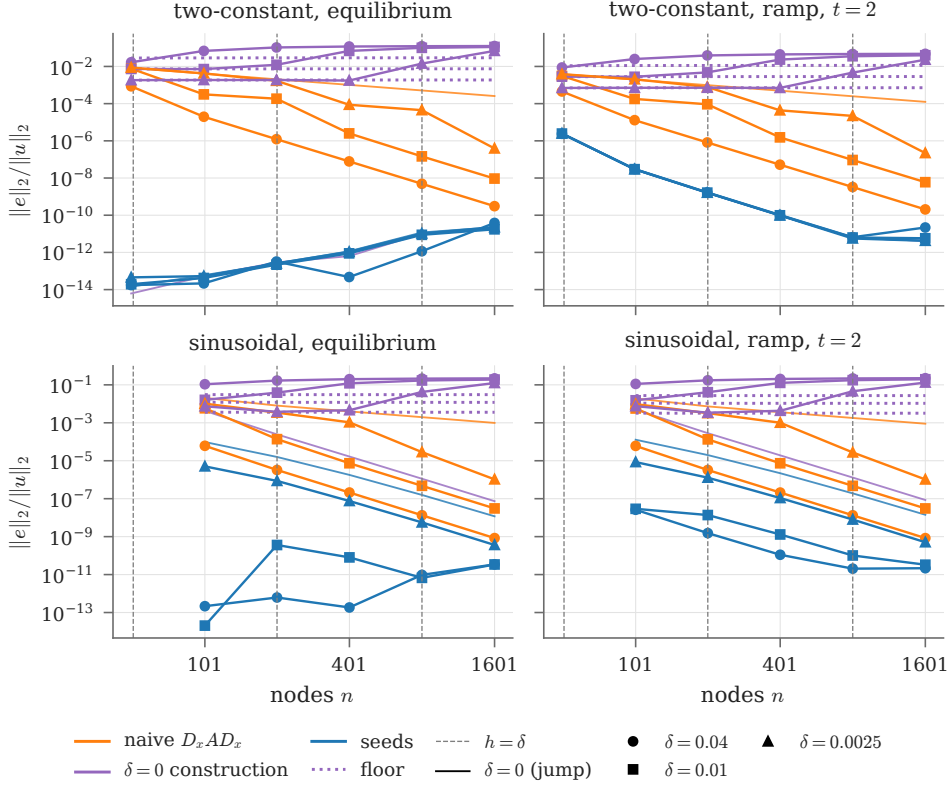


FIGURE 2. The naive operator $D_x A D_x$ (orange), the $\delta = 0$ construction (purple) and the seeds (blue) against the node count n on the two-constant medium (top) and the sinusoidal medium (bottom), at equilibrium (left) and on the ramp problem at $t = 2$ (right); errors are $\|e\|_2/\|u\|_2$ at the nodes. Markers give δ , and lines without markers are the jump, $\delta = 0$. Dotted: the floor $\|u_0 - u_\delta\|_2/\|u_\delta\|_2$ of each $\delta > 0$. Dashed verticals: $h = \delta$. At equilibrium on the two-constant medium the seeds, and at $\delta = 0$ the construction, are exact, and their lines are the direct solve's rounding.

to 4.23 up to the largest count. The equilibrium has the same knee at the same counts: the knee belongs to the operator, not to the problem.

4.3. The jump construction on a smooth edge. The $\delta = 0$ construction converges to the jump problem's solution, so while it converges the floor bounds its error from below. It has two regimes (Figure 2, purple, and Table 1).

Unresolved. While the grid does not resolve the edge, the construction sits on the floor. At $\delta = 0.0025$ on the two-constant medium its equilibrium error at 50, 100 and 200 nodes is the floor to three digits, and wherever $h \geq 2\delta$

TABLE 1. The naive operator and the $\delta = 0$ construction on the ramp problem at $t = 2$: $\|e\|_2/\|u\|_2$ at the nodes, against the Chebyshev reference at each δ . *floor*: the two references' difference $\|u_0 - u_\delta\|_2/\|u_\delta\|_2$ at the largest count. Last line: the constant c of the resistance deficit (5), measured from the two quadratures at every δ and in closed form.

n	naive $D_x A D_x$			$\delta = 0$ construction		
	$\delta = 0.04$	$\delta = 0.01$	$\delta = 0.0025$	$\delta = 0.04$	$\delta = 0.01$	$\delta = 0.0025$
<i>two-constant medium 1/9 1, ramp problem at $t = 2$</i>						
50	4.50×10^{-4}	3.09×10^{-3}	3.85×10^{-3}	8.80×10^{-3}	2.82×10^{-3}	6.97×10^{-4}
100	1.29×10^{-5}	1.79×10^{-4}	2.04×10^{-3}	2.51×10^{-2}	2.78×10^{-3}	7.12×10^{-4}
200	8.19×10^{-7}	9.25×10^{-5}	7.90×10^{-4}	3.91×10^{-2}	4.79×10^{-3}	7.16×10^{-4}
400	5.22×10^{-8}	1.54×10^{-6}	4.36×10^{-5}	4.45×10^{-2}	2.34×10^{-2}	7.01×10^{-4}
800	3.28×10^{-9}	9.32×10^{-8}	2.19×10^{-5}	4.67×10^{-2}	3.54×10^{-2}	4.65×10^{-3}
1600	2.05×10^{-10}	5.96×10^{-9}	2.20×10^{-7}	4.77×10^{-2}	3.99×10^{-2}	2.30×10^{-2}
floor				1.14×10^{-2}	2.87×10^{-3}	7.18×10^{-4}
<i>sinusoidal medium, ramp problem at $t = 2$</i>						
101	6.06×10^{-5}	5.67×10^{-3}	9.61×10^{-3}	1.12×10^{-1}	1.53×10^{-2}	7.51×10^{-3}
201	3.25×10^{-6}	1.34×10^{-4}	3.23×10^{-3}	1.73×10^{-1}	4.09×10^{-2}	3.41×10^{-3}
401	2.11×10^{-7}	7.36×10^{-6}	1.01×10^{-3}	2.04×10^{-1}	1.27×10^{-1}	4.29×10^{-3}
801	1.33×10^{-8}	4.76×10^{-7}	2.74×10^{-5}	2.18×10^{-1}	1.77×10^{-1}	4.60×10^{-2}
1601	8.36×10^{-10}	3.06×10^{-8}	1.03×10^{-6}	2.24×10^{-1}	1.97×10^{-1}	1.32×10^{-1}
floor				2.71×10^{-2}	1.06×10^{-2}	3.20×10^{-3}
two-constant medium: deficit $c\delta$, $c = 8.78890$ measured, 8.78890 closed form						

on that medium it lies at 0.93 to 1.00 times the floor, at equilibrium and on the ramp problem. The floor is first order in δ , 0.73 to 0.75δ at equilibrium and 0.29δ on the ramp problem at $t = 2$, since the edge's resistance falls short of the jump's by $c\delta$ with $c = 8.79$ (5), which the two quadratures reproduce to six digits at every width; the flux through the medium changes by $c\delta/F_0 = 0.88\delta$ relative to the jump's, $F_0 = 10$ being the jump's total resistance. On the sinusoidal medium the floor is not yet proportional to δ at these widths: it is 0.76 , 1.20 and 1.44δ at equilibrium for $\delta = 0.04$, 0.01 and 0.0025 , and the measured deficit is 0.44 , 0.74 and 0.91 of the limit 20.7δ of its two $10 : 1$ edges. The sinusoid has $\alpha'/\alpha = 25$ where it meets the edges, so that at $\delta = 0.01$ it changes by 75% within 3δ of the edge, and the pieces the edge blends are not the constants that (5) assumes. There the coarse rows also carry the sinusoid's own error, and the construction lies at 1.05 to 2.3 times the floor where $h \geq 2\delta$.

Resolved. Once the grid resolves the edge the construction leaves the floor and its error grows. Its plain rows are then fourth-order rows on a smooth coefficient, but its three or four rebuilt rows still impose the translated basis, a kink of slope ratio α^+/α^- at the edge's centre, on a solution whose slope hardly changes across the centre cell, and the discrete solution takes the

TABLE 2. The seeds on the ramp problem at $t = 2$: $\|e\|_2/\|u\|_2$ at the nodes, with the rate per doubling of n in parentheses. Errors of a few times 10^{-12} on the two-constant medium and a few times 10^{-11} on the sinusoidal one are the references' own floor (Section 4.1), where the rates mean nothing.

n	$\delta = 0$	$\delta = 0.04$	$\delta = 0.01$	$\delta = 0.0025$
<i>two-constant medium $1/9 1$, ramp problem, error (rate)</i>				
50	2.49×10^{-6}	2.39×10^{-6}	2.46×10^{-6}	2.48×10^{-6}
100	2.95×10^{-8} (6.40)	3.05×10^{-8} (6.29)	2.99×10^{-8} (6.36)	2.95×10^{-8} (6.39)
200	1.61×10^{-9} (4.19)	1.69×10^{-9} (4.17)	1.64×10^{-9} (4.19)	1.62×10^{-9} (4.19)
400	9.76×10^{-11} (4.05)	1.03×10^{-10} (4.04)	9.89×10^{-11} (4.05)	9.74×10^{-11} (4.05)
800	6.11×10^{-12} (4.00)	6.54×10^{-12} (3.97)	6.32×10^{-12} (3.97)	5.61×10^{-12} (4.12)
1600	4.18×10^{-12} (0.55)	2.18×10^{-11} (-1.74)	5.76×10^{-12} (0.13)	4.22×10^{-12} (0.41)
<i>sinusoidal medium, ramp problem, error (rate)</i>				
101	1.28×10^{-4}	2.61×10^{-8}	2.92×10^{-8}	8.50×10^{-6}
201	1.99×10^{-5} (2.68)	1.53×10^{-9} (4.09)	1.37×10^{-8} (1.09)	1.28×10^{-6} (2.73)
401	2.20×10^{-6} (3.18)	1.11×10^{-10} (3.79)	1.30×10^{-9} (3.40)	1.08×10^{-7} (3.57)
801	1.91×10^{-7} (3.53)	2.05×10^{-11} (2.43)	1.02×10^{-10} (3.68)	7.95×10^{-9} (3.77)
1601	1.42×10^{-8} (3.75)	2.16×10^{-11} (-0.08)	3.31×10^{-11} (1.62)	5.06×10^{-10} (3.97)

kink. On the two-constant medium the error then follows h/δ , not δ : at $h \approx \delta/2$ it is 0.0685 to 0.0687 for all three widths, which is 2.4, 9.2 and 37 times the floor at $\delta = 0.04$, 0.01 and 0.0025. At 1600 nodes it reaches 0.11 at equilibrium for $\delta = 0.01$, and for $\delta = 0.04$ it is 0.125 at equilibrium and 0.048 on the ramp problem (0.22 for both on the sinusoidal medium). Between the two regimes, at $h \approx 2\delta$, the partly resolved plain rows pull the solution toward the smooth one, and the error dips just below the floor, to 0.93 of it at $\delta = 0.0025$.

So treating the edge as a jump is the right baseline exactly where the grid cannot see the edge, and the worst of the three operators where it can. Using it means knowing δ , switching it off near $h = 2\delta$, and accepting a floor that falls only as δ . At $\delta = 0$ it is the interface-aware operator of [Mar16], exact at equilibrium on the two-constant medium and fourth order on the sinusoidal one (rates 3.88 to 4.12).

4.4. The seeds. *The two-constant medium.* Table 2 and the blue lines of Figure 2 give the seeds. On the two-constant medium their error is one line for every δ . At $\delta = 0$ they are the interface-aware operator (Section 3.2). On the ramp problem their errors at $\delta = 0.0025$ lie within 0.1 to 0.3% of the $\delta = 0$ line at every count to 400, at $\delta = 0.01$ within 1 to 1.5%, and at $\delta = 0.04$, where $\delta \approx h$ at 50 nodes and the smooth solution is a different function, within 3 to 5%. The rates at $\delta = 0$ are 6.4, 4.2 and 4.05, then 4.0 to 4.1 at every width, from 2.5×10^{-6} at 50 nodes to 6×10^{-12} at 800, where

every width reaches the references' floor. At equilibrium the seeds are exact at every width (Remark 4.1).

Time and space. These are the errors of the fully discrete scheme at $\Delta t = h$, and part of each is the time integrator's. With $\Delta t = h/8$ the spatial error alone at $\delta = 0$ is 1.47×10^{-8} , 1.05×10^{-9} , 7.40×10^{-11} and 4.5×10^{-12} at 100 to 800 nodes, rates 3.8, 3.8 and 4.0, with the $\delta = 0.0025$ line within 0.3% of it and the $\delta = 0.01$ line within 1.8%. The fourth order and the one line are the space discretisation's. The time error is about half the total at 100 nodes and a quarter from 200, and it is where the first rate of 6.4 comes from.

A resolved edge costs nothing. Across the knee, where the naive operator changes order and the construction leaves its floor, the seeds' line does not move: it is the same fourth-order line at every h/δ from 16 down to $1/8$, below which the references' floor takes over, with no threshold in δ (Section 3.1). At 200 nodes and $\delta = 0.0025$, where $h = 4\delta$, the seeds' error on the ramp problem is 1.6×10^{-9} , against the naive operator's 7.9×10^{-4} and the construction's 7.2×10^{-4} . The construction's floor falls as δ , so the seeds' margin over it is set by δ and closes at $\delta = 0$, where on constant pieces the two are one operator.

The sinusoidal medium. At equilibrium the seeded rows are exact on the sinusoidal medium too, and the seed operator is exact only where the seeded rows cover the layer. On both problems the rows whose stencils lie farther than 20δ from both edges are standard rows on the sinusoid, and their error is the line. At $\delta = 0.04$ the reach covers the layer, every row is seeded, and the ramp problem's error is 2.6×10^{-8} , 1.5×10^{-9} and 1.1×10^{-10} at 101, 201 and 401 nodes (rates 4.1 and 3.8), and then the references' floor. At $\delta = 0.01$ the reach covers the layer at 101 nodes only; from 201 on it leaves the rows near $x = 0.25$ standard, which carry 1.4×10^{-8} at 201 nodes, and the line is fourth order past them. With every row seeded instead, the $\delta = 0.01$ errors at 201 nodes are 1.3×10^{-13} at equilibrium and 1.7×10^{-9} on the ramp problem; the reach is the operator's one parameter, and we left it at 20δ . At $\delta = 0.0025$ and at the jump the interior of the layer is standard at every count, and the line is the sinusoid's own error under the standard fourth-order rows, which is still pre-asymptotic, the rows beside the edges seeing $h\alpha'/\alpha = 0.5$ at 101 nodes. At $\delta = 0$ it falls from 1.3×10^{-4} to 1.4×10^{-8} over 101 to 1601 nodes at rates rising from 2.7 to 3.75, and at $\delta = 0.0025$ its rates rise from 2.7 to 4.0. At $\delta = 0$ that line is still 6 to 38 times below the construction's (Section 3.2).

Stability. The seed operator's interior spectrum, with the Dirichlet rows removed as BD4 steps it, is real and negative on both media at $\delta = 0$ and 0.0025, and its extreme eigenvalue is the standard stencil's $-16/(3h^2)$ in the material with $\alpha = 1$, to 0.6% at 50 nodes. On the sinusoidal medium at 49 and 53 nodes, where $h\alpha'/\alpha \approx 1$ at the layer's edges, the seed operator is stable, with largest eigenvalue -2.05 , while the $\delta = 0$ construction has

eigenvalues at +248 and +8.8; the sweep starts at 101 nodes so that all three operators share its grids.

4.5. Coefficient treatments. The standing alternative to a new stencil changes the coefficient and keeps the scheme (Section 1.1). We ran four such treatments on the grids and problems above, against the same references:

- T1:** the harmonic mean $(b - a) / \int_a^b d\xi / \alpha$ of α over a window $[a, b]$ of one or two cells centred on each node, clipped at the ends of the domain and sampled into A in $D_x A D_x$;
- T2:** the arithmetic mean over one cell, used the same way;
- T0:** the edge widened to $\max(\delta, mh)$, $m = 1$ or 2 , a coefficient regularised on the scale of the grid [TE03];
- T1-FV:** the conservative three-point scheme with the face conductances (13).

T1-FV is Pan, Xu and Li’s harmonic average method [PXL26], exact at the one-dimensional equilibrium (Section 3.5). T1 is the like-for-like changed medium: a harmonic cell mean, which as a face conductance makes the conservative scheme exact at equilibrium, sampled at the nodes of the fourth-order operator, where it does not; we found no source that uses it there. The harmonic mean is the exact conductance between blocks of uniform conductivity, and may be inferior to other means where the conductivity varies smoothly between nodes [GA92], as it does here once $\delta > 0$. A fifth candidate, the band-limited coefficient of the seismic literature [Mit17, KWR22], was not found in use for diffusion and is left out. The comparison is against coefficient sampling on the same nodes and against our implementation of the cell harmonic and arithmetic means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, on one contrast and one test problem per dimension.

The ranking. Figure 3 gives every treatment on the ramp problem, and Table 3 the errors at $\delta = 0.0025$ in the order of the ranking. On the ramp problem of the two-constant medium, at every width while $h \geq 4\delta$, the jump included, the order in $\|e\|_2 / \|u\|_2$ is

$$\begin{aligned} \text{seeds} < \text{T1-FV} < \text{T1 (two cells)} < \text{naive} \gtrsim \text{T1 (one cell)} \\ &\approx \text{T2} < \text{T0} (m = 1) < \text{T0} (m = 2), \end{aligned}$$

with the middle three at most 2.02 times apart; at 200 nodes and $\delta = 0.0025$ the errors are 1.6×10^{-9} , 8.0×10^{-6} , 8.9×10^{-5} , 7.9×10^{-4} , 4.0×10^{-4} , 4.6×10^{-4} , 2.2×10^{-3} and 5.0×10^{-3} . At the jump the seeds lead T1-FV by 53 times at 50 nodes and by 2.1×10^4 at 400. On the sinusoidal medium, whose edges fall on nodes, a one-cell window straddles each edge, and at the jump the one-cell harmonic mean is 9 to 35 times below the naive operator. There T1-FV is ahead of the seeds at $\delta = 0$ below 801 nodes and at $\delta = 0.0025$ at 101, where the seeds’ line is the sinusoid’s pre-asymptotic

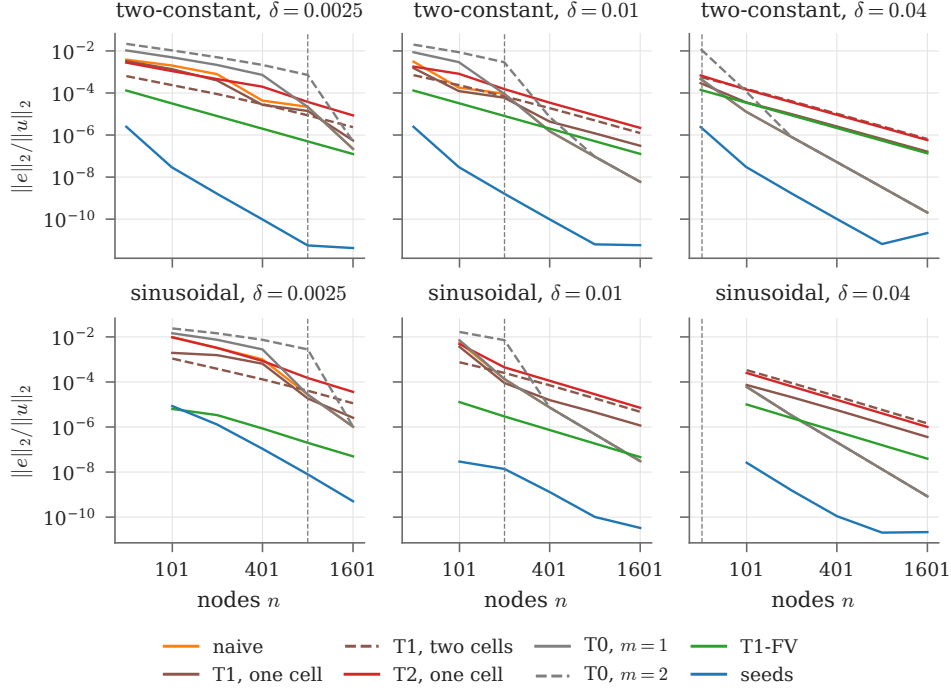


FIGURE 3. The coefficient treatments on the ramp problem at $t = 2$: $\|e\|_2/\|u\|_2$ at the nodes against n on the two-constant medium (top) and the sinusoidal medium (bottom) at $\delta = 0.0025, 0.01$ and 0.04 . The naive operator and the seeds are those of Figure 2, and the treatments those of Table 3. Dashed verticals: $h = \delta$. The seeds' flat ends are the references' floor.

one (Section 4.4) rather than the edge's; where the seeded rows cover the layer the seeds lead, by 440 times at 101 nodes and $\delta = 0.01$.

Second order. T1-FV is second order at every width, as Pan, Xu and Li prove, and on the two-constant medium with one constant: at $\delta = 0$ its error falls from 1.31×10^{-4} at 50 nodes to 1.26×10^{-7} at 1600, rate 2.00; at $\delta = 0.01$ and 0.0025 its errors lie within 1.2% of these at every count, and at $\delta = 0.04$ they lie 7% above. It is the strongest of the treatments. Every nodal treatment caps the fourth-order operator at second order once the grid resolves the edge: for $h \leq \delta/4$ on the two-constant medium the naive operator is fourth order and the treated ones second order (rates 1.95 to 2.00). At $\delta = 0.04$ and 1600 nodes the naive error is 2.05×10^{-10} , and the treated errors are 1.62×10^{-7} (T1, one cell), 5.76×10^{-7} (T2) and 6.50×10^{-7} (T1, two cells), 1.2 to 4.8 times T1-FV's 1.35×10^{-7} . A cell mean perturbs a smooth α at second order in the width of its window, and the fourth-order operator solves the perturbed problem faithfully. For the wave

TABLE 3. The treatments on the ramp problem at $t = 2$ and $\delta = 0.0025$: $\|e\|_2/\|u\|_2$ at the nodes, to two figures, in the order of the ranking of Section 4.5. T1: the harmonic mean over one or two cells; T2: the arithmetic mean over one cell; T0: the edge widened to $\max(\delta, mh)$; T1-FV: the conservative scheme with the face conductances (13).

$\ e\ _2/\ u\ _2$						
<i>two-constant medium 1/9 1, ramp problem, $\delta = 0.0025$</i>						
n	50	100	200	400	800	1600
seeds	2.5×10^{-6}	3.0×10^{-8}	1.6×10^{-9}	9.7×10^{-11}	5.6×10^{-12}	4.2×10^{-12}
T1-FV	1.3×10^{-4}	3.2×10^{-5}	8.0×10^{-6}	2.0×10^{-6}	5.0×10^{-7}	1.3×10^{-7}
T1, two cells	6.4×10^{-4}	2.4×10^{-4}	8.9×10^{-5}	3.0×10^{-5}	8.7×10^{-6}	2.4×10^{-6}
naive	3.8×10^{-3}	2.0×10^{-3}	7.9×10^{-4}	4.4×10^{-5}	2.2×10^{-5}	2.2×10^{-7}
T1, one cell	3.3×10^{-3}	1.4×10^{-3}	4.0×10^{-4}	2.7×10^{-5}	1.4×10^{-5}	5.5×10^{-7}
T2, one cell	2.8×10^{-3}	1.1×10^{-3}	4.6×10^{-4}	2.0×10^{-4}	3.8×10^{-5}	8.6×10^{-6}
T0, $m = 1$	1.1×10^{-2}	5.0×10^{-3}	2.2×10^{-3}	7.2×10^{-4}	2.2×10^{-5}	2.2×10^{-7}
T0, $m = 2$	2.2×10^{-2}	1.1×10^{-2}	5.0×10^{-3}	2.2×10^{-3}	7.2×10^{-4}	5.0×10^{-7}
<i>sinusoidal medium, ramp problem, $\delta = 0.0025$</i>						
n	101	201	401	801	1601	
seeds	8.5×10^{-6}	1.3×10^{-6}	1.1×10^{-7}	8.0×10^{-9}	5.1×10^{-10}	
T1-FV	6.4×10^{-6}	3.4×10^{-6}	8.6×10^{-7}	2.0×10^{-7}	5.0×10^{-8}	
T1, two cells	1.1×10^{-3}	3.8×10^{-4}	1.3×10^{-4}	4.1×10^{-5}	1.1×10^{-5}	
naive	9.6×10^{-3}	3.2×10^{-3}	1.0×10^{-3}	2.7×10^{-5}	1.0×10^{-6}	
T1, one cell	2.0×10^{-3}	1.6×10^{-3}	6.5×10^{-4}	1.9×10^{-5}	2.6×10^{-6}	
T2, one cell	9.7×10^{-3}	3.2×10^{-3}	8.6×10^{-4}	1.5×10^{-4}	3.6×10^{-5}	
T0, $m = 1$	1.5×10^{-2}	7.4×10^{-3}	2.8×10^{-3}	2.7×10^{-5}	1.0×10^{-6}	
T0, $m = 2$	2.4×10^{-2}	1.4×10^{-2}	7.4×10^{-3}	2.8×10^{-3}	1.0×10^{-6}	

equation with regularised discontinuities, Tornberg and Engquist report the same split: a second-order error component from the discontinuities under a higher-order discretisation [TE06]. So the nodal treatments share the construction's property: they help only while $h \gtrsim \delta$, and using them means knowing δ and switching them off, which the seeds need not.

The widened edge and the local defect. T0 is the worst. The widened medium's equilibrium differs from the true one by a resistance deficit $c(mh - \delta)$, and T0's equilibrium error is that difference, to 0.1% for $m = 2$ wherever $h \geq \delta$; on the two-constant medium's ramp problem at $\delta = 0$ it is 2.9 ($m = 1$) and 5.7 to 5.8 ($m = 2$) times the naive operator's error at every count. Widening an edge that the grid already fails to resolve adds resistance error and leaves the knee where it was. With the edge midway between nodes, a one-cell window ends on the jump at $\delta = 0$, and T1 (one cell) and T2 are the naive operator there to rounding. T1 over two cells converges at order 1.5 in this norm (rates 1.44 to 1.50 at $\delta = 0$ on the two-constant medium's ramp problem), the mark of a local defect: its error is first order on the two nodes beside the edge, 1.18×10^{-3} to 7.28×10^{-5} in the maximum norm

TABLE 4. The elliptic problem: $\|e\|_2/\|u\|_2$ at the nodes for the naive operator and the $\delta = 0$ construction at $\delta = 0.01$, the seeds at three widths, and T1-FV at $\delta = 0$ (Remark 4.1).

n	$\delta = 0.01$		seeds			T1-FV
	naive	construction	$\delta = 0$	$\delta = 0.01$	$\delta = 0.0025$	$\delta = 0$
<i>two-constant medium</i> $1/9 \mid 1$, <i>equilibrium</i> , $\ e\ _2/\ u\ _2$						
50	7.64×10^{-3}	7.35×10^{-3}	1.4×10^{-14}	1.9×10^{-14}	4.5×10^{-14}	2.7×10^{-14}
100	3.15×10^{-4}	7.21×10^{-3}	5.9×10^{-14}	4.2×10^{-14}	5.3×10^{-14}	8.2×10^{-15}
200	1.84×10^{-4}	1.23×10^{-2}	2.6×10^{-13}	2.2×10^{-13}	2.6×10^{-13}	1.2×10^{-13}
400	2.51×10^{-6}	6.86×10^{-2}	1.2×10^{-12}	8.9×10^{-13}	1.1×10^{-12}	2.1×10^{-13}
800	1.46×10^{-7}	1.00×10^{-1}	1.2×10^{-11}	8.8×10^{-12}	1.0×10^{-11}	1.7×10^{-12}
1600	9.34×10^{-9}	1.12×10^{-1}	2.3×10^{-11}	1.7×10^{-11}	2.3×10^{-11}	6.5×10^{-12}
<i>sinusoidal medium</i> , <i>equilibrium</i> , $\ e\ _2/\ u\ _2$						
101	5.83×10^{-3}	1.62×10^{-2}	9.7×10^{-5}	2.0×10^{-14}	5.0×10^{-6}	3.0×10^{-14}
201	1.38×10^{-4}	3.89×10^{-2}	1.6×10^{-5}	3.6×10^{-10}	8.6×10^{-7}	1.3×10^{-13}
401	7.38×10^{-6}	1.21×10^{-1}	1.8×10^{-6}	8.0×10^{-11}	7.5×10^{-8}	1.6×10^{-13}
801	4.77×10^{-7}	1.70×10^{-1}	1.6×10^{-7}	6.7×10^{-12}	5.6×10^{-9}	2.2×10^{-12}
1601	3.07×10^{-8}	1.90×10^{-1}	1.2×10^{-8}	3.4×10^{-11}	3.6×10^{-10}	1.4×10^{-11}

from 100 to 1600 nodes, and far smaller elsewhere, so that the relative ℓ^2 error goes as $h n^{-1/2}$. In the maximum norm it is first order.

Remark 4.1 (The elliptic problem). The left column of Figure 2 and Table 4 are the sanity check and the limit case, not a ranking (Section 3.5). On the two-constant medium the seeds are exact at every width, 1.9×10^{-14} at 50 nodes and $\delta = 0.01$, growing with n to 1.7×10^{-11} at 1600 through the direct solve's conditioning, as the interface-aware operator's own line at $\delta = 0$ does; T1-FV is exact too, 10^{-14} to 6.5×10^{-12} . The naive operator shows the knee of the ramp problem, from 7.6×10^{-3} at $h = 4\delta$ to 9.3×10^{-9} at $h = \delta/8$ for $\delta = 0.01$, and the construction the same floor. On the sinusoidal medium the equilibrium does rank the two, for a reason that has nothing to do with the edge. T1-FV stays exact, 3×10^{-14} to 1.4×10^{-11} at $\delta = 0$, while the seed operator is exact only where its seeded rows cover the layer, at $\delta = 0.04$ and, at 101 nodes, $\delta = 0.01$; elsewhere its error is that of the standard rows inside the layer, 9.7×10^{-5} to 1.2×10^{-8} at $\delta = 0$ over 101 to 1601 nodes and 5.0×10^{-6} to 3.6×10^{-10} at $\delta = 0.0025$.

4.6. A snapshot. Figure 4 and Table 5 put the four operators on one grid: the ramp problem at $t = 2$ on 100 nodes of the two-constant medium with $\delta = 0.0025$, so that $h = 8.08\delta$. It is the row of the sweep at which the four lines are ordered and apart: the naive operator at 2.04×10^{-3} , the construction 2.9 times below it, T1-FV 22 times below that, and the seeds 1100 times below that, at 2.95×10^{-8} . The naive operator's error is largest beside the edge, $1.5h$ to its left, and carries a node-to-node oscillation, the checkerboard mode of $D_x A D_x$ excited by the sampled edge; but 84% of $\|e\|_2^2$ lies more than $2h$ from the edge, on the side where $\alpha = 1/9$ and the profile

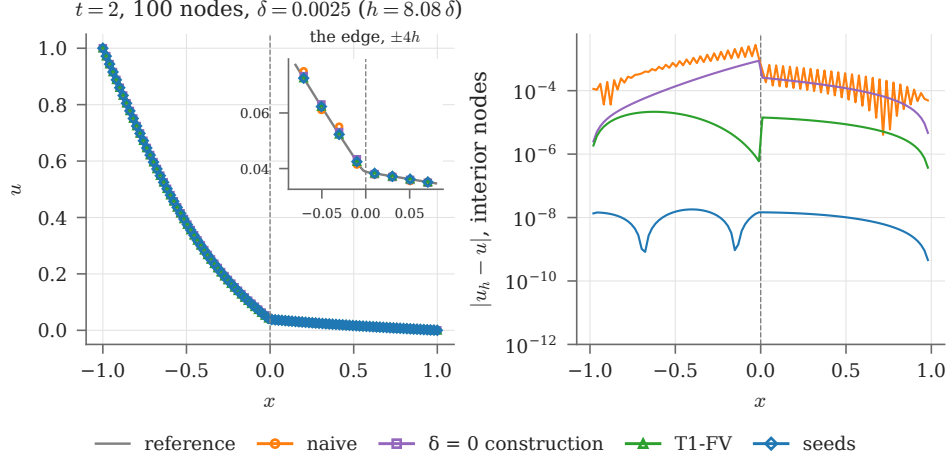


FIGURE 4. The ramp problem at $t = 2$ on 100 nodes of the two-constant medium with $\delta = 0.0025$ ($h = 8.08\delta$). Left: the reference and the nodal solutions of the four operators, with the edge enlarged over $\pm 4h$. Right: the pointwise error $|u_h - u|$ at the interior nodes. Dashed: the edge's centre.

TABLE 5. The snapshot of Figure 4: $\|e\|_2/\|u\|_2$ at the nodes, the largest nodal error and where it lies, and the share of $\|e\|_2^2$ from the nodes within $2h$ of the edge.

operator	$\ e\ _2/\ u\ _2$	$\max e $	at x	share within $2h$
naive $D_x A D_x$	2.04×10^{-3}	2.78×10^{-3}	-0.030	0.16
$\delta = 0$ construction	7.12×10^{-4}	8.79×10^{-4}	-0.010	0.23
T1-FV	3.20×10^{-5}	2.16×10^{-5}	-0.616	0.03
seeds	2.95×10^{-8}	1.80×10^{-8}	-0.394	0.07

is steep. The construction's error is smooth and global, the profile shifted by the resistance deficit, and largest at the node half a cell from the edge. T1-FV's is largest far from the edge, at $x = -0.62$: the face conductances take the edge exactly, and what remains is second-order truncation on the curved profile. The seeds' error is the $\delta = 0$ line's at this count, flat across the domain but for two sign changes, and about half of it is BD4's: with $\Delta t = h/8$ it is 1.47×10^{-8} .

5. SEEDS IN TWO DIMENSIONS

5.1. A straight feature. [*Stub, #48: the local frame and the anchor; seeds polynomial in the tangential coordinate with coefficient functions marched along the normal.*]

5.2. Stencil assembly. *[Stub, #48: the seeds in the RBF-FD system, the warp they bring with them, which rows are seeded; the seed-section figure.]*

5.3. Curved and tangentially varying features. *[Stub, #48: why a frozen normal profile is inconsistent where α varies along the edge; the tangential chain in the curve's own coordinates.]*

5.4. A thin resistive layer. *[Stub, #48: the double edge of a ring: its exact width, series composition, the flux seeds, the warp on rows anchored inside the layer.]*

6. TWO-DIMENSIONAL RESULTS

6.1. Test problems and references. *[Stub, #49: the flat band, the sine pair, the ring; the separable, product-grid and reference-free references (stiff 4.1, 4.6, 4.8; statement 1).]*

6.2. Naive baseline. *[Stub, #49: the knee on scattered nodes and the jump construction on its floor (stiff 4.2; statements 2–3).]*

6.3. Solvability and spectra. *[Stub, #49: the seed rows in the global matrix (stiff 4.4; statement 6).]*

6.4. The flat sweep. *[Stub, #49: the seeds against the baselines at every (n, δ) , the warp, the elliptic ranking (stiff 4.5; statements 4, 5, 7).]*

6.5. The curved feature. *[Stub, #49: the frozen profile and the tangential chain on the sine pair (stiff 4.6–4.7; statement 8).]*

6.6. The ring and the extremizing sweep. *[Stub, #49: the twins of EABE Fig. 19–20 and the smooth ring (stiff 4.8, 4.10; statement 9).]*

6.7. Coefficient treatments. *[Stub, #49: the treatments on scattered nodes (stiff 4.9; statement 10).]*

6.8. A snapshot. *[Stub, #49: the four regimes on one node set (stiff 5.1; statement 11).]*

7. DISCUSSION

7.1. Limitations. *[Stub, #50: corners, extreme contrasts, the cost in three dimensions, what the experiments leave open.]*

7.2. Future work. *[Stub, #50: in the order to do it.]*

7.3. Conclusions. *[Stub, #50: LITERATURE.md section 6b, verbatim.]*

ACKNOWLEDGMENTS

[Stub, #52: acknowledgments; where the code and data live.]

TOOL AND COMPUTATIONAL RESOURCE DISCLOSURE

[Stub, #52.]

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